

RATHOD AKASH

Hyderabad, India

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LinkedIn — GitHub

Professional Summary

Third-year Statistics and Data Science student at the Indian Statistical Institute, Bangalore, with hands-on experience in Python, machine learning, and data analysis. Skilled in building predictive models and deep learning systems, including MLPs and learning CNNs, with a focus on real world applications. Eager to leverage analytical and problem-solving skills in a Data Science or AI/ML internship.

Technical Skills

- **Programming:** Python, R, SQL-DBMS, DSA
- **Machine Learning:** Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Deployment
- **Deep Learning:** Neural Networks, CNN, MLP, NLP
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, OpenCV-ComputerVision
- **Data Analysis:** EDA, Statistical Analysis, Hypothesis Testing, Data Visualization, Text Preprocessing
- **Tools:** Jupyter Notebook, Git, GitHub, VS Code, Excel, FastAPI

Education

Bachelor of Statistics and Data Science

Indian Statistical Institute, Bangalore

Aug 2024 – Jun 2027

Intermediate (PCM) - 95.8%

TTWRS SOE, Khammam

Jun 2021 – May 2023

Schooling - 100% (10th Boards)

TSWRIES Junior College Boys, Kasipet

2011 – 2021

Project Experience

Zero-Trust Multi-Agent Trading Desk - (Google Vibe Coding Kaggle Capstone Project) -Link

July 2026

- Developed a secure Zero-Trust Multi-Agent AI Trading System using Python and FastAPI..
- Implemented deterministic policy enforcement, consensus validation, and Human-in-the-Loop approval for safe AI-assisted trading.
- Integrated cryptographic SHA-256 verification, Pydantic schema validation, and automated testing with 50+ test cases.
- Python, FastAPI, AsyncIO, Pydantic, Pytest, YAML, Multi-Agent Systems.

Forecast Accuracy Assessment – Teleflex (Assessment Project) -Link

Apr 2026

- Analyzed supply chain forecasts comparing Stat FC vs Cons FC using WAPE, MAE, and Bias.
- Performed segmentation (Category, Region, Promotion, Stockout) to identify key error drivers.
- Identified issues like promotion-driven demand spikes and stockout distortion affecting accuracy. s.
- Applied bias correction and promotion uplift techniques.
- Built a Machine Learning regression model reducing WAPE from 0.64 to 0.39 (38% improvement).
- Visualized insights using Excel (Pivot, Conditional Formatting) and Python (EDA, Seaborn).

Machine Learning Intern Projects – Unified Mentor -Link

Jun 2025 – Aug 2025

- Heart Disease Prediction: Applied Logistic Regression and Random Forest to predict heart disease from health metrics. Achieved 92
- Animal Classification using CNN models.
- Mobile Phone Price Prediction using classification techniques. Mobile Phone Price Prediction: Built classifiers to predict price categories based on specifications.
- Vehicle Price: Prediction: Developed regression models to forecast resale vehicle prices.
- Stock Price: Prediction using regression and visualization.
- Movie Recommender System: Developed vectorization-based similarity model.

Academic Projects

2025-2026

1. Neural ODE for Accuracy Prediction -Link

Working-2026

- Implemented Neural ODEs for modeling continuous-time data.
- Compared performance with RNN/LSTM models.
- Used PyTorch and ODE solvers for training and evaluation.

2. Counterfactual Fairness Through Data Orthogonalization -Link

June-2026

- Implemented fairness-aware preprocessing algorithms (OB, SOB) for counterfactual fairness in machine learning.
- Evaluated fairness using CF-Metric, Equal Opportunity, and Average Odds on Adult, COMPAS, and Loan datasets.
- Developed an Adaptive Fairness State (AFS-SOB) extension for continual fairness adaptation.
- Achieved 85.64% accuracy and 0.9045 ROC-AUC while reducing bias in sensitive attributes.

3. Metropolis Algorithm -Link

May-2026

- Developed a comprehensive Metropolis Algorithm simulation project using Markov Chain Monte Carlo (MCMC) methods in R. convergence diagnostics, burn-in analysis, and autocorrelation analysis for Markov chains.
- Applied Jackknife error estimation and statistical simulations for performance evaluation and validation.

4. Optimal Membership Taxation -Link

Optimization Project-2025

- Developed a convex optimization model for optimal taxation and welfare maximization. Simulated household income distributions using statistical modeling and real-world data.
- Implemented tax optimization and consumption analysis using CVXPY and Python.
- Visualized taxation patterns, spending behavior, and social welfare trade-offs.

Internships

Machine Learning Intern – Unified Mentor -Link

Jun 2025 – Aug 2025

- Worked on classification, regression, and visualization tasks using Python.

Campus Ambassador – Jyesta Corporation -Link

Jun 2025 – Jul 2025

- Promoted corporate initiatives and student engagement.

Additional Information

- **Languages:** English, Telugu, Hindi, Lambadi
- Open to relocation and remote opportunities